

Overlapping Generations: Foundations

Theory, Applications & Computation

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`https://alessiarussoecon.weebly.com/overlapping-generations-theory-applications--computation.html`

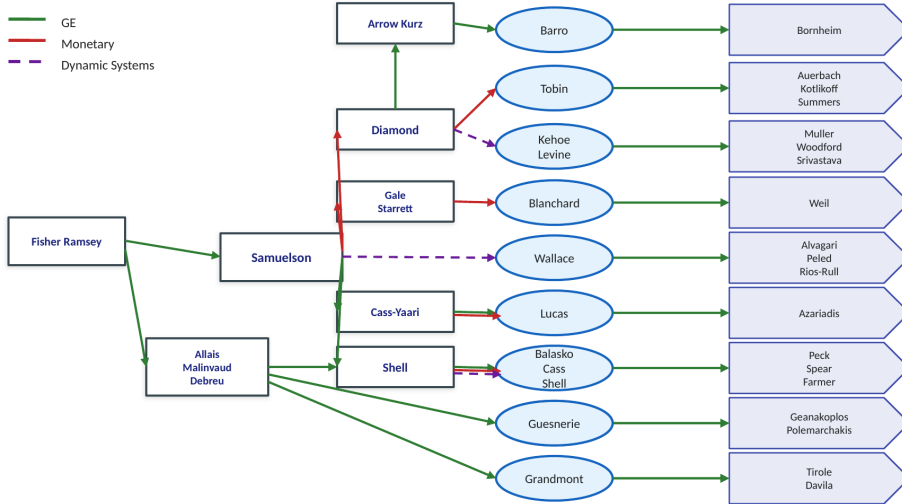
Why Overlapping Generations?

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Key applications

- Pensions and social security
- Public debt and fiscal policy
- Demographics and long-run growth
- Human capital accumulation
- Family economics and bequests
- Climate and environmental policy

OLG Morphology



Why Overlapping Generations?

Reference: S. Spear & W. Young, *Overlapping Generations: Methods, Models and Morphology* (Oxford University Press, 2023)

The OLG framework is the *natural* model of an economy in which

- agents are **finitely lived** and enter/exit the economy continuously,
- **intergenerational trade** and redistribution are possible (and welfare-relevant),
- **money, debt, and Social Security** have a meaningful role to play,
- markets may be **sequentially incomplete** and equilibria **dynamically inefficient**.

Yet from the 1980s onward OLG was **displaced** as the workhorse of macroeconomics by the infinitely-lived representative-agent (DSGE) paradigm (a limiting case of OLG where arrival rate of new agents is zero) — a framework that **rules out by assumption** many of the phenomena OLG was built to study.

A Brief History

- Allais (1947)** *Économie et Intérêt*: first formal OLG model of capital and interest; largely unrecognised in Anglophone literature for a decade.
- Samuelson (1958)** “An Exact Consumption-Loan Model...”: fiat money as a *social contrivance* sustaining Pareto-improving trade between generations; seeds of dynamic inefficiency and the Aaron condition.
- Shell (1971)** Introduces *sunspot equilibria* and equilibrium indeterminacy — phenomena impossible in Arrow–Debreu models with a finite horizon.
- Gale (1973)** Offer-curve geometry; distinction between the *Samuelson* (monetary) and *classical* equilibria; rigorous treatment of dynamic (in)efficiency.
- Barro (1974)** Ricardian Equivalence via *dynastic altruism*: if households care about their descendants, finite lives “unravel” — effectively collapsing OLG into a representative-agent model.

The Rise of a Dominant Paradigm

Following Barro (1974) and the Real Business Cycle programme (Kydland & Prescott, 1982), macroeconomics converged on the **DSGE framework**:

- A single **infinitely-lived representative household** — no birth, no death, no generational conflict.
- Complete markets, **Ricardian equivalence** holds: government debt is not net wealth.
- By construction, **no role for Social Security, pension reform, or demographic change** to affect aggregate welfare.

What is assumed away matters.

- Dynamic inefficiency $\Rightarrow$ impossible (transversality $\Rightarrow$ efficiency).
- Intergenerational redistribution $\Rightarrow$ neutral.
- Indeterminacy and sunspots $\Rightarrow$ excluded.

OLG vs. Ramsey–Cass–Koopmans

	Ramsey–CK	OLG
Agent horizon	Infinite	Finite
Heterogeneity	None	Overlapping generations
Ricardian equiv.	Holds	FAILS
Dynamic efficiency	Guaranteed	Not guaranteed
CE = planner opt.	Yes (1 st WT)	Not always
Sunspots / bubbles	No	Yes
Transversality cond.	Required	Not applicable

OLG and Heterogeneous-Agent Models

The OLG structure is a **natural prototype for heterogeneity**:

- Agents differ by **age** — and therefore by wealth, income, marginal propensities, and remaining horizon.
- Sequential market structure creates **endogenously incomplete markets**: the unborn cannot trade with the currently living.
- As a result, **government policy is non-neutral** — debt, Social Security, and taxation have genuine redistributive and welfare consequences *across generations*.

OLG thus offers a *disciplined* bridge between the representative-agent benchmark and the full heterogeneous-agent (HA-NK) literature: it isolates the **intergenerational** dimension of heterogeneity while remaining analytically tractable.

A Contemporary Revival

OLG models are **central once more**. Three forces drive the revival:

- **Demographic pressures** Ageing populations, falling fertility, and pension-system sustainability are first-order policy questions — questions DSGE cannot address.
- **Heterogeneity + redistribution** The HA-NK programme has made explicit the costs of abstracting from distributional dynamics; OLG provides the cleanest entry point.
- **Computational advances** Deep learning and global solution methods now make high-dimensional, multi-cohort models *feasible* — problems that were computationally intractable only a few years ago.

These lectures: back to basics.

We build the *core building blocks* — analytically and computationally — so that you can construct richer models and address relevant economic questions yourself.

The Ramsey Benchmark

The Ramsey Benchmark: Environment

Before introducing OLG formally, we establish the Ramsey model as a benchmark.

Primitives

- Single infinitely-lived representative agent
- Utility: $V = \sum_{t=1}^{\infty} \beta^{t-1} U(c_t)$, $\beta \in (0, 1)$
- Exogenous endowment $e_t > 0$ each period,
 $\sup_t e_t < \infty$
- U strictly concave, C^2 , Inada conditions
- **No production, no capital** — the only asset
is a one-period real bond a_t

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Two equivalent formulations

Arrow–Debreu (AD):

$$\max_{\{c_t\}} \sum_{t=1}^{\infty} \beta^{t-1} U(c_t) \quad \text{s.t.} \quad \sum_{t=1}^{\infty} p_t c_t \leq \underbrace{\sum_{t=1}^{\infty} p_t e_t}_W$$

Sequential:

$$a_{t+1} = (1 + r_t) a_t + e_t - c_t, \quad a_1 = 0$$

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Sequential:

$$a_{t+1} = (1 + r_t) a_t + e_t - c_t, \quad a_1 = 0$$

- The two formulations are equivalent *if and only if* the Transversality Condition (TVC) holds: $\lim_{T \rightarrow \infty} p_T a_{T+1} = 0$.
- Well-posedness of the AD problem requires $W < \infty$.

The Ramsey Benchmark: Equilibrium Characterisation

With a single agent and no storage, **goods clearing gives** $c_t = e_t$ **directly**. Prices are then residual.

FOC at date t : $\beta^{t-1} U'(c_t) = \lambda p_t$. Dividing by the date-1 FOC ($\lambda = U'(e_1)$, $p_1 = 1$):

Fundamental AD Pricing Formula

$$p_t = \beta^{t-1} \frac{U'(e_t)}{U'(e_1)} \iff \underbrace{\frac{p_t}{p_{t+1}}}_{\equiv 1 + r_{t+1}} = \frac{U'(e_t)}{\beta U'(e_{t+1})} \iff p_t = \prod_{s=2}^t (1 + r_s)^{-1}$$

Constant endowment $e_t = e$:

$$1 + r = \frac{1}{\beta} \implies r = \frac{1 - \beta}{\beta} > 0$$

The real interest rate is pinned *solely* by the discount factor β , independent of the level of endowment.

The Ramsey Benchmark: Equilibrium Characterisation

TVC. Iterate the flow constraint forward T periods:

$$\sum_{t=1}^T p_t c_t + p_T a_{T+1} = (1 + r_1) a_1 + \sum_{t=1}^T p_t e_t$$

As $T \rightarrow \infty$: the TVC $\lim_{T \rightarrow \infty} p_T a_{T+1} = 0$ collapses this to the AD budget constraint.

The Ramsey Benchmark: What It Delivers

Four Properties of the Ramsey Equilibrium

Property	Mechanism
Real rate pinned uniquely	$1 + r = 1/\beta$ from Euler eq.; β is a preference parameter, not an equilibrium object
Dynamic efficiency	$r = 1/\beta - 1 > 0 = n$ always; over-accumulation is impossible
TVC rules out non-optimal paths	Eliminates Ponzi schemes ($a_T \rightarrow -\infty$) <i>and</i> over-saving ($a_T \rightarrow +\infty$); equilibrium is unique
FWT holds	$W = \sum p_t e_t < \infty$ supports the CE as PO

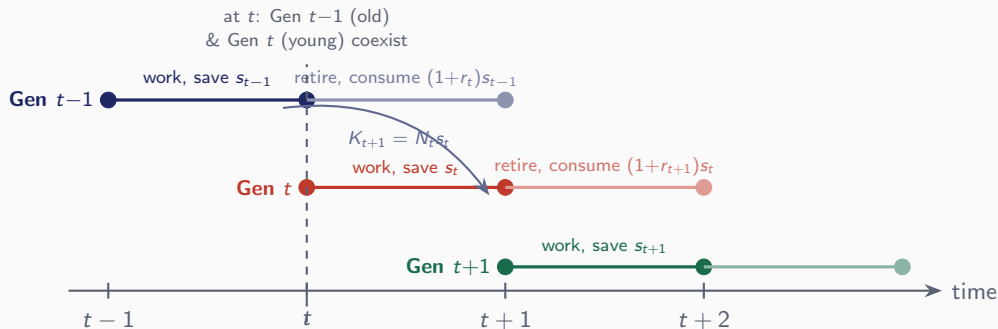
Ramsey vs. OLG: What Breaks and Why

What Breaks

Feature	Ramsey	OLG
Agent horizon	Infinite	Finite (2 periods)
TVC	Required, holds	Inapplicable
$\sum_t p_t e_t < \infty$	Guaranteed	Not guaranteed
FWT	Holds	May fail
Dynamic efficiency	Holds	May fail
Equilibrium	Unique	Multiple possible

Model Setup

OLG Timing: the Overlapping Structure



$N_t = (1+n)N_{t-1}$. At each date **two** generations coexist. Capital at $t+1$ equals *aggregate savings of the young* at t .

Basic Environment (Samuelson, 1958)

Time, goods, demographics

- Discrete time: $t = 1, 2, 3, \dots$
- Single non-storable consumption good (equivalently, net return to storage $r = -1$)
- No uncertainty, perfect foresight
- Each generation lives **exactly two periods**
- At date t : gen $t - 1$ (old) & gen t (young) coexist
- Initial old gen 0 lives only in period 1

Endowments (stationary): $e_t^t = w_1$, $e_{t+1}^t = w_2$.

Gen 0: endowment w_2 , outside money $M \geq 0$.

Preferences: $u_t = U(c_t^t) + \beta U(c_{t+1}^t)$,
 $\beta > 0$.

U : increasing, strictly concave, C^2 , Inada.

Key contrast with Ramsey

Each OLG budget involves *only two dates*:

$$p_t c_t^t + p_{t+1} c_{t+1}^t = p_t w_1 + p_{t+1} w_2$$

No TVC. No guarantee $\sum_t p_t \bar{e}_t < \infty$.

Timing Table

Gen \ Period	1	2	...	t	$t+1$
0	(c_1^0, e_1^0)				
1	(c_1^1, e_1^1)	(c_2^1, e_2^1)			
$\vdots$		$\ddots$			
$t-1$				(c_t^{t-1}, e_t^{t-1})	
t				(c_t^t, e_t^t)	(c_{t+1}^t, e_{t+1}^t)
$t+1$					$(c_{t+1}^{t+1}, e_{t+1}^{t+1})$

Each row = one generation's life; each column = one date. At date t , exactly **two** rows are non-empty.

Feasible allocation: $c_t^{t-1} + c_t^t = w_1 + w_2 \equiv W \ \forall t \geq 1$. **Pareto optimal:** no reallocation weakly improves everyone with at least one strict improvement.

Equilibrium Concepts

Two Equilibrium Concepts

Arrow–Debreu Equilibrium

All trading at $t = 1$. Prices $\{p_t\}_{t=1}^{\infty}$, $p_1 = 1$.

Generation $t \geq 1$ solves:

$$\max U(c_t^t) + \beta U(c_{t+1}^t)$$

$$\text{s.t. } p_t c_t^t + p_{t+1} c_{t+1}^t = p_t w_1 + p_{t+1} w_2$$

Euler equation:

$$\frac{p_t}{p_{t+1}} = \frac{U'(c_t^t)}{\beta U'(c_{t+1}^t)}$$

Goods market clearing ($\forall t$):

$$c_t^t + c_t^{t-1} = w_1 + w_2 = W$$

Sequential Markets Equilibrium

Spot markets each period. Rate r_{t+1} .

Generation $t \geq 1$ solves:

$$\max U(c_t^t) + \beta U(c_{t+1}^t)$$

$$\text{s.t. } c_t^t + s^t = w_1, \quad c_{t+1}^t = w_2 + (1 + r_{t+1})s^t$$

Euler equation:

$$U'(c_t^t) = \beta(1 + r_{t+1}) U'(c_{t+1}^t)$$

Good market clearing ($\forall t$):

$$c_t^t + c_t^{t-1} = w_1 + w_2 = W$$

$\Rightarrow$ **Asset market clearing ($\forall t$):** $s^t = \frac{M}{p_t}$

AD–SM Equivalence and the Source of Multiplicity

Proposition (equivalence)

AD $\Rightarrow$ SM: Set $1 + r_{t+1} = p_t/p_{t+1}$ and $s^t = w_1 - c_t^t$: SM budget constraints and Euler equation are satisfied.

SM $\Rightarrow$ AD: Normalise $p_1 = 1$; set $p_{t+1} = p_t/(1 + r_{t+1})$ recursively. The SM budget collapses to the AD budget constraint. $\square$

Two corollaries:

1. **AD prices:** $p_t = \prod_{\tau=2}^t (1 + r_\tau)^{-1}$ — same formula as Ramsey, but *no TVC* pins down the level.
2. **Real money demand (monetary equilibrium only):** asset clearing $s^t = M/p_t$ pins the real value of savings once $p_1 > 0$ is determined by expectations.

Source of Multiplicity

Ramsey: TVC $\lim_T p_T a_{T+1} = 0$ pins the unique real interest rate r (equivalently, the price ratio p_t/p_{t+1}) consistent with individual optimality and goods clearing.

OLG: No TVC; each generation's budget has *two terms* only. Multiple values of r are consistent with individual optimality and goods clearing simultaneously.

$\Rightarrow$ The real interest rate r is **not pinned down** by the equilibrium conditions alone.

The Samuelson Condition

The Autarkic Equilibrium

At autarky ($M = 0$), market clearing gives $c_t^t = w_1$, $c_{t+1}^t = w_2 \forall t$. The OLG Euler equation at the endowment pins down **autarkic AD prices**:

$$\frac{p_t}{p_{t+1}} = \frac{U'(w_1)}{\beta U'(w_2)} \equiv 1 + \bar{r} \quad \implies \quad p_t = (1 + \bar{r})^{-(t-1)}$$

Autarkic interest rate

$$1 + \bar{r} = \frac{U'(w_1)}{\beta U'(w_2)}$$

Agent's MRS at the endowment point = equilibrium relative price of date- t vs date- $(t + 1)$ goods = slope of the offer curve at autarky.

Gale's Classification: Samuelson vs. Classical

Gale (1973) classification

Samuelson: $\bar{r} < 0$, i.e. $U'(w_1) < \beta U'(w_2)$. Young *want to save* at $r = 0$.

Classical: $\bar{r} > 0$, i.e. $U'(w_1) > \beta U'(w_2)$. Young *do not want to save*.

Knife-edge: $\bar{r} = 0$.

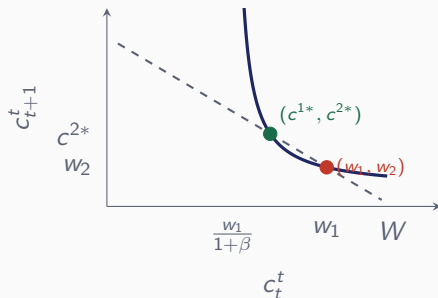
Marginal gain from saving at $r = 0$:

$$\left. \frac{du}{ds} \right|_{s=0} = -U'(w_1) + \beta U'(w_2) \begin{cases} > 0 & \text{Samuelson} \\ = 0 & \text{knife-edge} \\ < 0 & \text{Classical} \end{cases}$$

If $p_1 = 1$: $\bar{r} > 0 \Rightarrow \sum_t p_t W < \infty$;
 $\bar{r} \leq 0 \Rightarrow \sum_t p_t W = +\infty$.

Canonical Example

$U(c) = \log c$, $\beta = 0.75$, $w_1 = 1$, $w_2 = 0.3$:
 $\bar{r} = w_2/(\beta w_1) - 1 = -0.6$. **Samuelson.**



Why Is the Offer Curve Slope $-(1 + \bar{r})$ at the Endowment?

In (c_t^t, c_{t+1}^t) space, the offer curve passes through (w_1, w_2) .

Three-step argument

1. At autarky: $(c_t^t, c_{t+1}^t) = (w_1, w_2)$, agent at endowment.
2. FOC pins the supporting budget line: $q_{\text{aut}} = \beta U'(w_2)/U'(w_1) = 1/(1+\bar{r})$, so slope $= -(1+\bar{r})$.
3. Offer curve is the locus of tangencies, so its slope at (w_1, w_2) equals that of the supporting budget: $dc_{t+1}^t/dc_t^t|_{(w_1, w_2)} = -(1+\bar{r})$.

Implication (feasibility line always has slope -1):

- Samuelson ($\bar{r} < 0$): slope $-(1+\bar{r}) \in (-1, 0)$, *shallower* than -1 . Offer curve exits (w_1, w_2) **above** feasibility line.
- Classical ($\bar{r} > 0$): slope ≤ -1 , *steeper* than -1 . Offer curve exits (w_1, w_2) **below** feasibility line.

Offer Curves & Graphical Analysis

Two Representations of the Offer Curve

Consumption space (c_t^t, c_{t+1}^t)

Axes = observable consumptions.

Endowment (w_1, w_2) on curve. CE = curve $\cap$ feasibility line.

Feasibility: $c_t^t + c_{t+1}^t = w_1 + w_2$, slope -1 (*no prices*).

Budget slope = $-(1 + r_{t+1})$.

Excess demand space (y, z)

(Gale 1973). $y \equiv c_t^t - w_1$, $z \equiv c_{t+1}^t - w_2$.

Autarky = $(0, 0)$. Feasibility: $y + z = 0$, slope -1 (shifted).

Slope at origin = $-(1 + \bar{r})$.

Feasibility condition

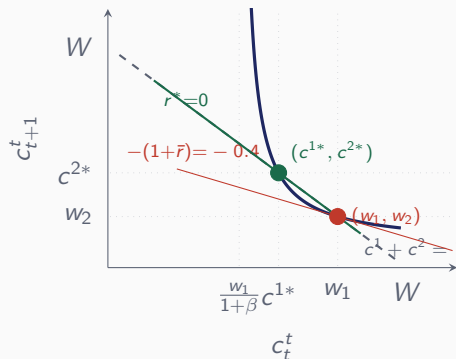
$c_t^{t-1} + c_t^t = w_1 + w_2$: non-storable good $\Rightarrow$ total consumption = endowment. **One unit less by the young = one unit more for the old.** Rate = -1 . Budget slope $-(1 + r)$ varies; feasibility slope is always -1 . They coincide only when $r = 0$.

The Offer Curve: Properties in Consumption Space

Log utility: $c_{t+1}^t = \frac{\beta w_2 c_t^t}{(1 + \beta)c_t^t - w_1}$ (rectangular hyperbola, asymptote at $c_t^t = w_1/(1 + \beta)$). **Key properties:**

1. (w_1, w_2) always on the curve
2. Slope at (w_1, w_2) : $-(1 + \bar{r})$
3. Samuelson: slope $\in (-1, 0)$, shallower than -1
4. Classical: slope ≤ -1 , steeper than -1

In (y, z) : same curve, autarky $= (0, 0)$, feasibility $y + z = 0$.



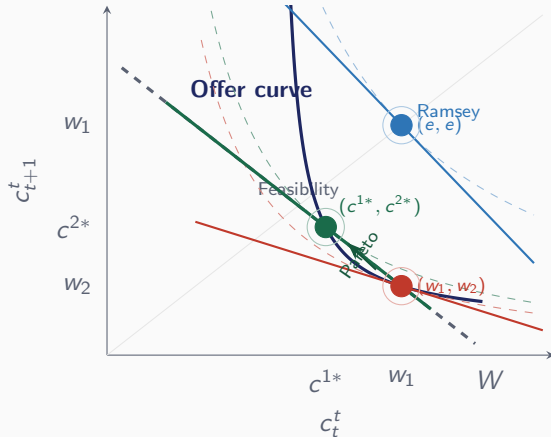
OLG and Ramsey on the Same Diagram

Three equilibria:

- **Red** (w_1, w_2) : OLG autarky. Slope $= -(1+\bar{r}) = -0.4$. Suboptimal.
- **Green** (c^{1*}, c^{2*}) : monetary eq. Slope $= -1$ ($r^*=0$). Budget $\equiv$ feasibility.
- **Blue** $(e, e) = (w_1, w_1)$: Ramsey. Slope $= -1/\beta = -4/3$. TVC $\Rightarrow$ FWT holds.

Slope comparison

Red (OLG aut.): $-0.4 > -1$ shallower
Grey (feasib.): -1
Blue (Ramsey): $-4/3 < -1$ steeper
Samuelson $\Leftrightarrow$ autarky budget shallower
than feasibility $\Leftrightarrow$ 2nd crossing exists.



Finding the Fixed Point i

Substitute the feasibility line $c^2 = W - c^1$ ($W \equiv w_1 + w_2$) into the offer curve

$$c^2 = \frac{\beta w_2 c^1}{(1 + \beta)c^1 - w_1}:$$

$$W - c^1 = \frac{\beta w_2 c^1}{(1 + \beta)c^1 - w_1}$$

Rearranging yields:

$$[(1 + \beta)c^1 - W][c^1 - w_1] = 0$$

Two fixed points and their c^2 from feasibility.

Finding the Fixed Point ii

Equilibrium allocations

	c^1	c^2
Autarky	w_1	w_2
Monetary eq.	$\frac{W}{1+\beta} = c^{1*}$	$\frac{\beta W}{1+\beta} = c^{2*}$

Canonical example ($\beta = 0.75$, $w_1 = 1$, $w_2 = 0.3$, $W = 1.3$):

$$c^{1*} = \frac{1.3}{1.75} = 0.743, \quad c^{2*} = \frac{0.75 \times 1.3}{1.75} = 0.557.$$

The Social Planner's Problem

A utilitarian planner assigns weight λ_t to generation t :

$$\max_{\{c_1^t, c_2^t\}} \sum_{t=1}^{\infty} \lambda_t \left[\log c_1^t + \beta \log c_2^t \right] \quad \text{s.t.} \quad c_1^t + c_2^{t-1} = W \quad \forall t.$$

FOCs (shadow price μ_t on the date- t resource constraint)

$$c_1^t : \frac{\lambda_t}{c_1^t} = \mu_t \qquad c_2^{t-1} : \frac{\lambda_{t-1} \beta}{c_2^{t-1}} = \mu_t$$

For a **stationary** solution ($c_1^t = c^1$, $c_2^t = c^2$, $\mu_t = \mu$ for all t), shadow prices are constant iff $\lambda_t = \lambda$:
equal weights across all generations.

Solution. Combine $c^2 = \beta c^1$ with feasibility $c^1 + c^2 = W$:

$$c^1(1 + \beta) = W \implies c^{1*} = \frac{W}{1 + \beta} = 0.743, \quad c^{2*} = \frac{\beta W}{1 + \beta} = 0.557.$$

The Social Planner's Solution: Equivalences

The planner's stationary optimum coincides with the monetary equilibrium

What the planner does that the market cannot

The feasibility constraint $c_1^t + c_2^{t-1} = W$ involves *two generations simultaneously*. The planner internalises the **intergenerational externality**: when generation t saves, it benefits generation $t - 1$ (the old), but the market provides no compensation for this transfer. The shadow price μ_t corrects the missing price.

Implementation: Second Welfare Theorem for OLG

Fiat money **decentralises** the planner's optimum: voluntary exchange between generations replicates the equal-weight allocation *without coercion*, provided the initial old (gen 0) are endowed with the money stock $M > 0$ (we see it later).

This is the OLG analogue of the Second Welfare Theorem: the constrained-optimal allocation is achievable as a CE with the right endowment of outside money.

The Offer Curve in (y, z) Space

$y \equiv c_t^t - w_1$, $z \equiv c_{t+1}^t - w_2$: shift origin to autarky.

Properties:

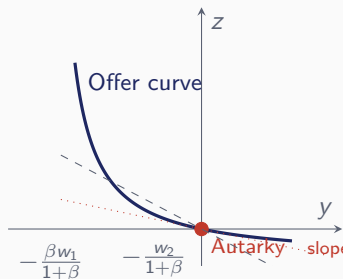
1. $y \geq -w_1$, $z \geq -w_2$ (non-negativity)
2. Degree-zero homogeneous in (p_t, p_{t+1})
3. Walras: $p_t y + p_{t+1} z = 0$, $z/y = -p_t/p_{t+1}$
4. Autarky: $(y, z) = (0, 0)$ always on curve
5. Slope at origin: $-(1 + \bar{r})$

Samuelson: *shallower* than $y + z = 0$ (slope > -1).

Classical: *steeper* (slope ≤ -1).

Log example ($\beta = 0.75$, $w_1 = 1$, $w_2 = 0.3$):

$$y(q) = \frac{0.3q - 0.75}{1.75}, \quad z(q) = \frac{0.75/q - 0.3}{1.75}.$$



Constructing the Offer Curve: Log Utility

$$U(c) = \log c, \beta = 0.75, w_1 = 1, w_2 = 0.3.$$

Derivation in consumption space

FOC: $U'(c_t^t) = \beta(1+r)U'(c_{t+1}^t)$, i.e. $c_{t+1}^t = \beta(1+r)c_t^t$.

Budget ($q = 1/(1+r)$): $c_t^t + qc_{t+1}^t = w_1 + qw_2$. Solve:

$$c_t^t(q) = \frac{w_1 + qw_2}{1 + \beta}, \quad c_{t+1}^t(q) = \frac{\beta c_t^t(q)}{q}.$$

Eliminating q : $c_{t+1}^t = \frac{\beta w_2 c_t^t}{(1 + \beta)c_t^t - w_1} = \frac{0.225 c_t^t}{1.75 c_t^t - 1}$.

In (y, z) : $s(q) = (\beta w_1 - q w_2)/(1 + \beta)$, $y = -s$, $z = s/q$.

Key values

$$q_{\text{aut}} = \beta w_1 / w_2 = 2.5 \quad (\bar{r} = -0.6)$$

At $r^* = 0$ ($q = 1$):

$$c^{1*} = 1.3/1.75 = 0.743$$

$$c^{2*} = 0.75 \cdot 1.3/1.75 = 0.557$$

$$\text{Check: } c^{1*} + c^{2*} = 1.3 = w_1 + w_2 \quad \checkmark$$

High q (future costly) $\Rightarrow$ save less
Low q (future cheap) $\Rightarrow$ save more
Varying q traces the curve.

Finding Equilibria: The Cobweb Algorithm

CE = offer curve $\cap$ feasibility line. $W \equiv w_1 + w_2$; $c_2^0 \equiv w_2 + z_0$ (gen-0 old); $z_0 \equiv m/p_1 > 0$.

Steps in consumption space

Step 1. Goods clearing: $c_1^1 = W - c_2^0 = w_1 - z_0$.

Step 2. Vertical: read c_2^1 from offer curve at c_1^1 (gen 1 optimises).

Step 3. Horizontal: clearing $c_1^2 = W - c_2^1$; read c_2^2 ; iterate.

Every $z_0 \in (0, w_1)$ gives a **distinct** CE path.

Canonical example: $U(c) = \log c$, $\beta = 0.75$, $w_1 = 1$, $w_2 = 0.3$, $\bar{r} = w_2/(\beta w_1) - 1 = -0.6$; $z_0 = 0.15$:

t	c_1^t	c_2^t	$y_t = c_1^t - w_1$	$z_t = c_2^t - w_2$
1	0.850	0.392	-0.150	+0.092
2	0.908	0.347	-0.092	+0.047
3	0.953	0.321	-0.047	+0.021

Path converges to **autarky** $(w_1, w_2) = (1.0, 0.30)$. Monetary eq. $(c^{1*}, c^{2*}) = (0.743, 0.557)$ is the *geometric intersection* of offer curve and feasibility line — an unstable fixed point; not a limit of iteration.

The Cobweb Map: Fixed Points and Stability i

Define $T(c_t^1) \equiv W - \text{offer}(c_t^1)$, so $c_{t+1}^1 = T(c_t^1)$. Stationary CE $\Leftrightarrow \underbrace{c^2 = \text{offer}(c^1)}_{\text{optimality}}$ and

$$\underbrace{c^1 + c^2 = W}_{\text{feasibility}} \Leftrightarrow T(c^1) = c^1.$$

Step 1 — Closed form of T . Substitute $\text{offer}(c^1) = \frac{\beta w_2 c^1}{(1 + \beta)c^1 - w_1}$:

$$T(c^1) = W - \frac{\beta w_2 c^1}{(1 + \beta)c^1 - w_1} = \frac{W[(1 + \beta)c^1 - w_1] - \beta w_2 c^1}{(1 + \beta)c^1 - w_1}$$

Using $W = w_1 + w_2$, the numerator becomes $[(1 + \beta)w_1 + w_2]c^1 - w_1 W$:

$$T(c^1) = \frac{[(1 + \beta)w_1 + w_2]c^1 - w_1 W}{(1 + \beta)c^1 - w_1}$$

The Cobweb Map: Fixed Points and Stability ii

Step 2 — Fixed points. Solve $T(c^1) = c^1$:

$$[(1 + \beta)w_1 + w_2]c^1 - w_1W = c^1[(1 + \beta)c^1 - w_1]$$

Rearranging:

$$(1 + \beta)(c^1)^2 - [(1 + \beta)w_1 + W]c^1 + w_1W = 0$$

This factors *exactly* as:

$$[(1 + \beta)c^1 - W][c^1 - w_1] = 0$$

giving exactly two stationary CE:

$$c^1 = w_1 \text{ (autarky)} \quad \text{and} \quad c^1 = \frac{W}{1 + \beta} = c^{1*} \text{ (monetary eq.)}$$

For the canonical example: $c^{1*} = 1.3/1.75 = 0.743$, $c^{2*} = \beta W/(1 + \beta) = 0.557$.

The Cobweb Map: Fixed Points and Stability iii

Step 3 — Derivative and stability.

$$\text{offer}'(c^1) = \frac{d}{dc^1} \left[\frac{\beta w_2 c^1}{(1 + \beta)c^1 - w_1} \right] = \frac{-\beta w_1 w_2}{[(1 + \beta)c^1 - w_1]^2} < 0$$

$$\Rightarrow T'(c^1) = -\text{offer}'(c^1) = \frac{+\beta w_1 w_2}{[(1 + \beta)c^1 - w_1]^2} > 0$$

Evaluating at each fixed point (denom. = βw_1 at autarky, = w_2 at monetary eq.):

Stability

Autarky: $T'(w_1) = \frac{w_2}{\beta w_1} = 1 + \bar{r} = 0.4 < 1$ **stable.**

Monetary eq.: $T'(c^{1*}) = \frac{\beta w_1}{w_2} = 2.5 > 1$ **unstable.**

Key chain: autarky stable $\Leftrightarrow T'(w_1) < 1 \Leftrightarrow 1 + \bar{r} < 1 \Leftrightarrow \bar{r} < 0 \Leftrightarrow \beta > w_2/w_1$ (Samuelson).

The Intuition behind Unravelling

Starting with $c_1^1 = 0.85 > c^{1*} = 0.743$: generation 1 saves *less* than the monetary equilibrium requires ($z_1 = 0.15 < z^* = 0.257$).

The feedback mechanism

1. Low savings by gen $t \Rightarrow$ old gen t receives few resources $\Rightarrow c_2^t$ low.
2. Goods clearing: $c_1^{t+1} = W - c_2^t$ large $\Rightarrow$ gen $t+1$ consumes more when young, saves even less.
3. Lower savings $\Rightarrow$ lower real return to saving next period $\Rightarrow$ gen $t+2$ saves less still.
4. $z_t \rightarrow 0$: the economy unravels to autarky.

The real return along the path falls monotonically:

	Monetary eq.	$t = 1$	$t = 2$	$t = 3$	Autarky
c_1^t	0.743	0.850	0.908	0.953	1.000
c_2^t	0.557	0.392	0.347	0.321	0.300
$1 + r$	1.000	0.615	0.510	0.449	0.400

The Intuition behind Unravelling

Why the monetary equilibrium is not the attractor

The monetary equilibrium requires **knife-edge coordination**: every generation must believe the next will accept money at value z^* . If gen 1 saves slightly less, money is worth less, the real return falls, and gen 2 rationally saves less still — the expectation of monetary equilibrium **unravels**.

Autarky needs **no coordination**: if you expect the next generation will not accept money, your best response is also not to save. Autarky is *self-confirming*.

The tragedy: the *better* equilibrium (monetary) is fragile; the *worse* equilibrium (autarky) is robust. This is why social contrivances — PAYG, public debt, fiat money — are needed to coordinate on the Pareto-superior outcome.

Cycles in OLG

The shape of the offer curve beyond autarky depends on whether the **substitution** or **income** effect dominates a rise in r .

Forward-bending ($\sigma > 1$, EIS high)

Substitution effect $>$ income effect.

Savings **increase** as r rises.

Offer curve: as $c_t^t \downarrow$ (savings $\uparrow$), $c_{t+1}^t \uparrow$ monotonically.

Cobweb map T is **monotone**.

No cycles possible.

Log utility ($\sigma = 1$) is the knife-edge case between forward and backward bending.

Backward-bending ($\sigma < 1$, EIS low)

Income effect $>$ substitution effect.

Savings **decrease** as r rises past a threshold.

Offer curve *folds back*: c_{t+1}^t peaks then falls as c_t^t decreases.

Cobweb map T becomes **non-monotone**.

Cycles possible.

Pareto Inefficiency and the FWT

The First Welfare Theorem: Proof (1/2)

Proof by contradiction. Assume the CE $\{c_t^i\}$ with prices $\{p_t\}$ is *not* Pareto optimal: $\exists$ a feasible $\{\hat{c}_t^i\}$ that Pareto-dominates it.

Steps 1–2: individual budgets

1. Strict preference $\Rightarrow$ unaffordable.

By Pareto dominance, $\exists i^*$ who *strictly* prefers $\hat{c}^{i^*}$ over c^{i^*} . But c^{i^*} is already the *best affordable* bundle at prices $\{p_t\}$. If $\hat{c}^{i^*}$ were affordable, i^* would have chosen it — contradicting optimality of c^{i^*} . Hence $\hat{c}^{i^*}$ must exceed the budget:

$$\sum_t p_t \hat{c}_t^{i^*} > \sum_t p_t e_t^{i^*}.$$

2. Weak preference $\Rightarrow$ not strictly cheaper.

Every other agent i weakly prefers $\hat{c}^i$. By the same revealed-preference argument, $\hat{c}^i$ cannot be strictly cheaper than the endowment (otherwise c^i would not have been optimal):

$$\sum_t p_t \hat{c}_t^i \geq \sum_t p_t e_t^i \quad \forall i \neq i^*.$$

The First Welfare Theorem: Proof (2/2)

Steps 3–4: aggregation and contradiction

3. Aggregate: the strict inequality survives.

Sum steps 1–2 across all i . Since at least one inequality (for i^*) is strict:

$$\sum_i \sum_t p_t \hat{c}_t^i > \sum_i \sum_t p_t e_t^i.$$

Swap $\sum_i \leftrightarrow \sum_t$ — valid iff $\sum_t p_t \bar{e}_t < \infty$ (Fubini–Tonelli; this is what fails in OLG):

$$\sum_t p_t \sum_i \hat{c}_t^i > \sum_t p_t \sum_i e_t^i.$$

4. Feasibility $\Rightarrow$ contradiction.

$\{\hat{c}_t^i\}$ is feasible by assumption: $\sum_i \hat{c}_t^i \leq \sum_i e_t^i$ for all t . Since $p_t > 0$ for all t :

$$\sum_t p_t \sum_i \hat{c}_t^i \leq \sum_t p_t \sum_i e_t^i.$$

This contradicts step 3. The Pareto-dominating allocation cannot exist. □

Fubini–Tonelli: Where the Proof Breaks in OLG

Fubini–Tonelli Theorem

For *non-negative* $f_{i,t} \geq 0$: $\sum_i \sum_t f_{i,t} = \sum_t \sum_i f_{i,t}$ always holds.

For *signed* $f_{i,t}$: interchange valid **iff** $\sum_{i,t} |f_{i,t}| < \infty$ (absolute convergence required).

Step 3 requires swapping $\sum_i \leftrightarrow \sum_t$ in $\sum_i \sum_t p_t \hat{c}_t^i$. A sufficient condition is $\sum_t p_t \bar{e}_t < \infty$, where $\bar{e}_t \equiv \sum_i e_t^i$ (total endowment at t).

Ramsey: swap valid

Prices: $p_t = \beta^{t-1} U'(e)/U'(e_1) \rightarrow 0$.

$$\sum_{t=1}^{\infty} p_t \bar{e} = \frac{\bar{e}}{1-\beta} < \infty.$$

$\text{TVC} + \beta < 1$ guarantee convergence.

Swap valid $\Rightarrow$ **FWT holds**.

OLG Samuelson: swap fails

Prices at autarky: $p_t = (1+\bar{r})^{-(t-1)} \rightarrow +\infty$
since $1+\bar{r} \in (0, 1)$.

$$\sum_{t=1}^{\infty} p_t W = +\infty.$$

Absolute convergence fails.

Swap **invalid** $\Rightarrow$ **FWT can fail**.

How to Cure Inefficiency

Implementing the Pareto-Optimal Allocation

In the Samuelson OLG economy, the PO allocation requires an intergenerational transfer from young to old. This can be decentralised through **three equivalent institutions**:

Pension (PAYG)

Tax the young, pay the old each period.

Transfer enforced by the **government**.

Public Debt

Young buy bonds; old sell them to the next young.

Rolled over **forever** by the government.

Fiat Money

Young hold money; old sell it to the next young.

Sustained by **self-fulfilling beliefs**.

Equivalence

All three deliver the same real allocation: a perpetual transfer of resources from young to old. They differ only in the *institution* that makes the transfer credible — fiscal authority (pension, debt) vs. coordination on a *belief* that money retains value.

How to Reach a Pareto Optimal Equilibrium?

Common Logic

All three mechanisms implement the **same intergenerational transfer**. They differ in who enforces it (state vs. beliefs) and in their robustness to political risk, inflation, and sunspot volatility. We will study pension/debt mechanisms in the next lectures. Now we focus on the role of (fiat) money.

Monetary Equilibrium

Why Is Money Valued in Equilibrium?

*Why would rational agents pay real goods for intrinsically useless pieces of paper?
(Pieces of paper with pictures of dead president)*

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What “money” means here

- Fiat currency: paper with no intrinsic value
- Produced by the government at zero cost
- Yields no dividend, no coupon, no utility in itself
- Cannot be consumed or used in production

The puzzle

If money is useless and costless to produce, a rational agent should never sacrifice real goods to obtain it. Yet in every economy we observe it.

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Two answers

- **Money as a bubble.** Its value derives solely from a self-fulfilling expectation.

$$q_t = \frac{q_{t+1}}{1 + r_t} \Rightarrow q_t = \lim_{T \rightarrow \infty} \frac{q_T}{\prod_{s=1}^T (1 + r_s)}$$

- **Money as a rate-of-return asset.** It is held only if:

$$\frac{q_{t+1}}{q_t} \leq 1 + r_t$$

At an interior solution, money and bonds must yield the same return.

Why do agents hold money? The OLG/bubble approach is not the only answer.

1. Cash-in-Advance (CIA)

Idea: Money must be acquired before goods can be purchased — a technological constraint on transactions.

$$P_t c_t \leq M_t \quad (\text{CIA constraint})$$

Strength: Tractable; delivers a clear money demand and connects to quantity theory.

Limitation: The constraint is *ad hoc* — it imposes by assumption the very thing that needs to be explained. Why must money, specifically, be used?

2. Money in the Utility Function (MIU)

Idea: Real money balances M_t/P_t enter utility directly, capturing liquidity services in reduced form.

$$V = \sum_{t=1}^{\infty} \beta^{t-1} U\left(c_t, \frac{M_t}{P_t}\right)$$

Strength: Flexible and analytically convenient; nests CIA as a limiting case.

Limitation: A reduced form — it assumes the conclusion. The microeconomic reasons why liquidity generates utility are left unmodelled.

3. Search and Matching

Idea: Decentralised bilateral trade is costly; a universally accepted medium of exchange reduces search frictions. Which object becomes money is a *social convention* — an equilibrium selection.

Strength: Genuine microfoundation; explains *why* a medium of exchange is valuable and which objects are likely candidates (divisible, durable, recognisable).

Limitation: Does not cleanly explain why *fiat* money — intrinsically worthless — displaces commodity money or credit, nor why credit arrangements do not eliminate the need for money altogether.

Can money alleviate dynamic inefficiency?

- Previous models lacked a long-lived asset that would facilitate intergenerational trade.
- Money could solve this problem.

An OLG Model of Bubble Money

Start with the standard two-period OLG model **without** production or bonds.

Demographics & Preferences

Cohort size: $N_t = (1+n)^t$, $n \geq 0$

Utility: $u(c_t^t, c_{t+1}^t)$
strictly concave, C^2 ,
Inada conditions

Endowments: $w_1 > 0$ when young
 $w_2 \geq 0$ when old

Technology & Markets

Technology: Goods can only be consumed
— no storage, no capital.

Markets: goods (price P_t),
money (numéraire)

Goods Market Clearing

$$\underbrace{N_t c_t^t}_{\text{young consumption}} + \underbrace{N_{t-1} c_t^{t-1}}_{\text{old consumption}} = \underbrace{N_t w_1}_{\text{young endowment}} + \underbrace{N_{t-1} w_2}_{\text{old endowment}}$$

Introducing Money

Setup

- In period $t = 1$, the initial old are endowed with M_0 units of fiat money (“bits of green paper”).
- The government costlessly prints money each period at a **constant** gross growth rate:

$$\frac{M_{t+1}}{M_t} = 1 + \theta, \quad \theta \geq 0$$

- M_t is the total money stock entering period t .

Getting Money into Circulation

Problem: how does newly printed money reach households?

Money pays nominal interest.

- An old household who carries M_t units of money into period t receives a transfer of θM_t units from the government.
- θ is the nominal interest rate of holding money.
- *Note:* You can replace $M_t = B_t$ nominal bonds, and $\theta = i$ nominal interest rate on bonds.

Timing within Period t

Beginning of t

Young (generation t):

- N_t agents are born
- each receives endowment $w_1 > 0$
- born with no money holdings

Old (generation $t - 1$):

- N_{t-1} agents alive
- each receives endowment $w_2 \geq 0$
- carry over M_{t-1} from last period
- receive government transfer θM_{t-1}
- now hold $M_t = (1 + \theta)M_{t-1}$ in total

During t : Trade

- Young **sell goods** to the old
- Old **sell money** to the young
- Trade clears through the Walrasian auctioneer at price P_t
- Young pay $P_t x_t$ nominal dollars for x_t real units of saving
- Old use money to buy goods and consume c_t^{t-1}

Timing within Period t

End of t

The young hold M_t units of money, to be carried into period $t + 1$ as the old.

Key: this is the **intergenerational transfer** absent in the non-monetary model — the young sacrifice goods today in exchange for money they expect to spend tomorrow.

Household Budget Constraints

Young (generation t , born in period t)

Nominal form:

$$P_t w_1 = P_t c_t^t + P_t x_t$$

Real form:

$$w_1 = c_t^t + x_t$$

- w_1 : real endowment received when young
- c_t^t : real consumption when young
- $x_t = w_1 - c_t^t$: real saving (purchases of money)
- $P_t x_t$: nominal outlay on money

Old (generation t , in period $t + 1$)

Nominal form:

$$P_{t+1} c_{t+1}^t = P_{t+1} w_2 + x_t (1 + \theta) P_t$$

Real form:

$$c_{t+1}^t = w_2 + x_t R_{t+1}$$

- w_2 : real endowment received when old
- $x_t (1 + \theta) P_t$: nominal money payoff
- R_{t+1} : gross real return on money:

$$R_{t+1} = (1 + \theta) \frac{P_t}{P_{t+1}} = \frac{1 + \theta}{1 + \pi_{t+1}}$$

The Savings Function

Solving the optimization problem yields optimal saving as a function of the real return and endowments:

$$x_t = s(R_{t+1}, w_1, w_2)$$

Properties (general):

- $x_t \geq 0$ if $w_2 = 0$ (no old-age income $\Rightarrow$ must save to consume when old)
- $\partial s / \partial R_{t+1}$ **ambiguous** in general: income and substitution effects work in opposite directions

Log utility special case ($u = \log c_t^t + \beta \log c_{t+1}^t$):

$$x_t = \frac{\beta w_1}{1 + \beta} - \frac{w_2}{(1 + \beta)R_{t+1}}$$

- With $w_2 = 0$: $x_t = \beta w_1 / (1 + \beta)$ **independent of R_{t+1}**
- Income and substitution effects cancel exactly under log utility when $w_2 = 0$
- With $w_2 > 0$: $\partial s / \partial R > 0$ (substitution effect dominates)

General Point: Asset Neutrality

The Household Problem is Generic

It does not matter **what asset** the household saves in — money, bonds, capital, ... Without uncertainty, all assets must pay the **same gross real return** R_{t+1} in equilibrium. Otherwise arbitrage.

The **solution** $s(R_{t+1})$ is always the same. Only what R_{t+1} *is in equilibrium* changes across models.

What Changes Across Models

Asset	Gross real return	Determined by
Capital	$1 + f'(k_t)$	Production function (MPK)
Bonds	R_t	Asset market clearing
Money	$(1 + \theta) P_t / P_{t+1}$	Money market clearing + inflation

In this model, saving takes the form of **money purchases** ($x_t = M_t / (N_t P_t)$), but the household cares only about the implied R_{t+1} not about the physical form of the asset.

Equilibrium

Equilibrium: Markets and Clearing Conditions

Government Policy

The government is described solely by a **constant money growth rule**:

$$\frac{M_{t+1}}{M_t} = 1 + \theta, \quad \theta \geq 0$$

No taxes, no spending — money creation is the only fiscal instrument.

Goods Market Clearing

$$N_t c_t^t + N_{t-1} c_t^{t-1} = N_t w_1 + N_{t-1} w_2$$

Dividing by N_t :

$$c_t^t + \frac{c_t^{t-1}}{1+n} = w_1 + \frac{w_2}{1+n}$$

Money Market Clearing

The young buy money from the old:

$$M_t = N_t P_t x_t$$

In per-capita real terms:

$$m_t \equiv \frac{M_t}{N_t P_t} = x_t = s(R_{t+1})$$

M_t exogenous P_t adjusts to clear the market.

Competitive Equilibrium

A sequence $\{c_t^t, c_t^{t-1}, x_t, m_t, P_t\}_{t=0}^{\infty}$ constitutes a **competitive equilibrium** if:

1. **Household optimality:** each generation t chooses (c_t^t, c_t^{t+1}, x_t) to maximise $u(c_t^t, c_t^{t+1})$ subject to the two budget constraints, taking P_t, R_{t+1} as given.
2. **Government:** the money growth rule $M_{t+1}/M_t = 1 + \theta$ holds every period.
3. **Goods market clears** every period.
4. **Money market clears** every period: $M_t = N_t P_t x_t$.

Characterising Equilibrium: State Variables

Predetermined (carried into t):

- M_t : nominal money stock (set by last period's young)
- $N_t = (1 + n)^t$: cohort size (exogenous, but a state)

We track only the **ratio**:

$$m_t \equiv \frac{M_t}{N_t P_t}$$

What about P_t ?

- Normally: flexible prices adjust to clear markets and are not states.
- Here: P_t is determined by expectations of P_{t+1} — money is a bubble, so P_t is *forward-looking*.
- Tracking m_t absorbs both M_t and P_t and reduces the system to **one scalar**.

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Derive a **single difference equation** in m_t that characterises all equilibrium paths. Steady states of this equation are monetary equilibria.

Deriving the Law of Motion

Money Market Clearing

$$m_t = s(R_{t+1})$$

Express R_{t+1} in Terms of m

From the money growth rule:

$$1 + \theta = \frac{M_{t+1}}{M_t} = \frac{m_{t+1}}{m_t} \cdot \frac{P_{t+1}}{P_t} \cdot \frac{N_{t+1}}{N_t} = \frac{m_{t+1}}{m_t} \cdot \frac{P_{t+1}}{P_t} \cdot (1 + n)$$

Solving for inflation:

$$\frac{P_{t+1}}{P_t} = \frac{(1 + \theta)}{(1 + n)} \cdot \frac{m_t}{m_{t+1}}$$

Therefore the gross real return on money is:

$$R_{t+1} = (1 + \theta) \frac{P_t}{P_{t+1}} = (1 + n) \frac{m_{t+1}}{m_t}$$

Deriving the Law of Motion

Law of Motion

$$m_t = s \left((1 + n) \frac{m_{t+1}}{m_t} \right) \quad (2)$$

A single implicit equation in (m_t, m_{t+1}) . Higher $m_{t+1}/m_t \Rightarrow$ lower inflation $\Rightarrow$ higher real return $\Rightarrow$ higher saving.

Explicit Derivation: Separable Utility

Setup: Separable Utility

Assume $u(c_t^t, c_{t+1}^t) = v(c_t^t) + \beta v(c_{t+1}^t)$ with v strictly concave.

Substituting into the Euler Equation

The Euler equation $v'(c_t^t) = \beta R_{t+1} v'(c_{t+1}^t)$, with $c_t^t = w_1 - x_t$ and $c_{t+1}^t = w_2 + R_{t+1} x_t$, becomes:

$$v'(w_1 - x_t) = \beta R_{t+1} v'(w_2 + R_{t+1} x_t)$$

Now substitute $x_t = m_t$ and $R_{t+1} = (1 + n) m_{t+1} / m_t$:

$$v'(w_1 - m_t) = \beta (1 + n) \frac{m_{t+1}}{m_t} \cdot v'(w_2 + (1 + n) m_{t+1})$$

From Offer Curve to Law of Motion

Reinterpreting the Axes

Horizontal axis = m_t

Money market clearing gives:

$$m_t = x_t = w_1 - c_t^t$$

Distance from endowment to optimal c_t^t
equals real per-capita money demand m_t .

Therefore the offer curve implicitly defines:

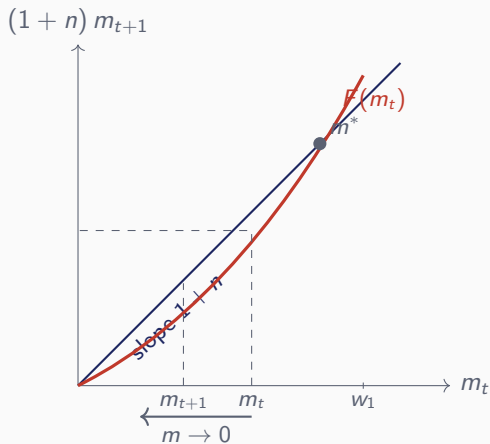
$$(1 + n) m_{t+1} = F(m_t)$$

Vertical axis = $(1 + n)m_{t+1}$

For the old generation:

$$\begin{aligned} c_{t+1}^t - w_2 &= R_{t+1} s(R_{t+1}) \\ &= (1 + n) \frac{m_{t+1}}{m_t} \cdot m_t \\ &= (1 + n) m_{t+1} \end{aligned}$$

Law of Motion: Phase Diagram



Steady state m^* :

- Intersection of $F(m)$ and slope- $(1+n)$ ray.
- Unique (when offer curve not backward bending).
- **Unstable**: any $m_0 \neq m^*$ leads away.

Off-steady-state:

- $m_0 > m^*$: infeasible (consumption exceeds output).
- $m_0 < m^*$: $m_t \rightarrow 0$ via accelerating inflation.

Steady State: Existence and Properties

The Monetary Steady State

Definition: $m_t = m_{t+1} = m^*$.

At the steady state the $(1 + n)$ ray intersects the offer curve:

$$(1 + n) m^* = F(m^*)$$

Unique when the offer curve is not backward bending.

It is unstable.

Golden Rule property:

Since $m_{t+1}/m_t = 1$ in steady state:

$$R = (1 + n) \frac{m^*}{m^*} = 1 + n$$

The real return on money equals the population growth rate — the **Golden Rule**.

Steady-state inflation:

$$\frac{P_{t+1}}{P_t} = \frac{1 + \theta}{1 + n}$$

Interpretation of the Golden Rule

$R = 1 + n$ means that every unit saved by a young agent of cohort t yields $(1 + n)$ units for the old, exactly matching the growth of the young cohort. Money is the vehicle that makes this intergenerational transfer feasible. If $\theta = n$, the price level is constant (zero inflation).

Dynamics: Non-Backward-Bending Case

Assumption: Offer Curve Not Backward Bending

Saving is non-decreasing in R : $\partial s / \partial R \geq 0$. This holds under log utility with $w_2 = 0$.

$m_0 > m^*$: Infeasible

- If $m_0 > m^*$, the map $(1+n)m_{t+1} = F(m_t)$ implies $m_t \rightarrow \infty$.
- But $m_t \leq w_1$ always (young cannot save more than their endowment).
- **No equilibrium with $m_0 > m^*$ exists.**

$m_0 < m^*$: Inflationary Collapse

- $m_t \rightarrow 0$: real value of money vanishes.
- Since M grows at rate θ , $m \rightarrow 0$ requires $P \rightarrow \infty$ (hyperinflation).
- As R falls over time, saving falls further $\Rightarrow$ inflation accelerates.
- **Self-fulfilling collapse** of the monetary equilibrium.

Intuition for the Instability

At m^* : agents save just enough to keep m constant. If $m_0 < m^*$: R is slightly lower $\Rightarrow$ agents save less $\Rightarrow m_1 < m_0 \Rightarrow R$ falls further $\Rightarrow \dots$ The monetary equilibrium unravels.

Dynamics: Backward-Bending Offer Curve

Initial Money Stock: Indeterminacy

What Pins Down m_0 ?

Nothing in the model pins down m_0 .

Any $m_0 \in (0, m^*)$ is consistent with equilibrium: the young are willing to hold that amount of money as long as they expect the price level tomorrow to make R_{t+1} consistent with their saving choice.

- There is a **continuum of equilibrium paths**, one for each $m_0 \in (0, m^*)$.
- $m_0 = 0$ (autarky) is also an equilibrium.

Why? Money is a bubble.

- Money has no intrinsic value.
- Its value today derives solely from the belief that it will have value tomorrow.
- Any $m_0 > 0$ consistent with that belief is an equilibrium — a self-fulfilling prophecy.
- This is the fundamental **indeterminacy** of fiat money in OLG models.

Dynamic Efficiency and Money

Can Money Solve Dynamic Inefficiency?

Money might, because it **permits intergenerational trade**: the young give up goods today in exchange for money they will spend when old. This is exactly the transfer that was missing in the non-monetary OLG model.

Whether it actually does depends on the **slope of the offer curve at the origin**.

Why the Slope at the Origin?

At the origin, $x_t = 0$: households neither borrow nor lend. This is the autarkic allocation — the **non-monetary equilibrium**.

The slope of the offer curve at (w_1, w_2) equals the autarkic real interest rate R^A .

Comparing R^A to $1 + n$ tells us whether the non-monetary economy is dyn. efficient.

Two Cases

Samuelson

OFC flatter than
 $1 + n$ at origin

$$R^A < 1 + n$$

Dyn. inefficient

Monetary eq. **exists**

Classical

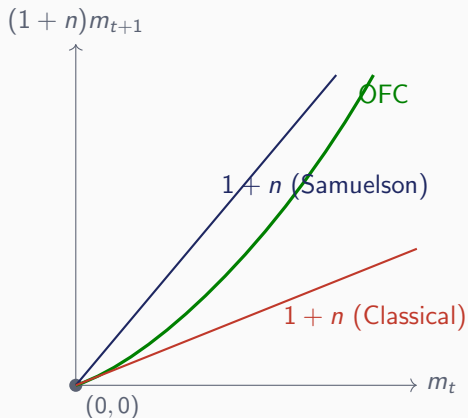
OFC steeper than
 $1 + n$

$$R^A > 1 + n$$

Dyn. efficient

Monetary eq. **does not exist**

Samuelson and Classical Cases: Diagram



Samuelson case:

- OFC flatter than $1+n$ ray at origin.
- At $R = 1+n$, households *want* to save $\Rightarrow$ monetary equilibrium exists.
- $m^* > 0$ at the intersection.

Classical case:

- OFC steeper than $1+n$ ray.
- No intersection for $m > 0$.
- Monetary equilibrium does not exist.

Main Result on Money and Dynamic Efficiency

Result

*Money is valued in equilibrium **only** in an economy that would be dynamically inefficient without money.*

Samuelson Case: Money Helps

- Without money: $R^A < 1 + n$. Economy over-accumulates; young save too much.
- With money: $R = 1 + n$ at the monetary steady state. The Golden Rule is attained.
- Money **improves welfare** by providing a store of value with return $1 + n > R^A$.

Classical Case: Money Irrelevant

- Without money: $R^A > 1 + n$. Economy is already efficient.
- Households would not hold money at $R = 1 + n < R^A$: real assets dominate.
- No monetary equilibrium with $m > 0$ exists.
- Same logic rules out welfare-improving PAYG pensions (Aaron condition $r > n$).

Is This a Good Theory of Money?

Good Features

1. **Money valued endogenously.** Its value is not assumed (no utility from money, no CIA constraint) — it emerges from equilibrium.
2. **Fragility is a feature.** The value of money depends entirely on expectations. Confidence crises and hyperinflation are natural outcomes, not anomalies.

Limitations

1. **No rate-of-return dominance.** In the model, money and bonds earn the same real return. In reality, money pays zero nominal interest and is dominated by short bonds — yet people hold it.
2. **No liquidity.** The model has no frictions, no transactions costs, no reason to prefer money over other assets for making payments.

Open Question

How to construct a theory of money that resolves both limitations without introducing new *ad hoc* assumptions is an **open research question** in monetary economics.

Take the classical OLG indeterminacy and ask:
can fiscal policy act as an equilibrium-selection device?

Paper Money, Sims (2013)

In the Samuelson pure exchange OLG with outside money M :

Monetary steady state

$P_{t+1}/P_t = 1/(1+n)$; money valued as a bubble;
Pareto-improves on autarky in the Samuelson case.

Hyperinflationary continuum

$M/P_t \rightarrow 0$ along a one-parameter family of paths indexed by P_0 . Money eventually becomes worthless.

+ *the autarky equilibrium, where money is **never** valued.*

The OLG fragility problem

The “good” monetary equilibrium is **not selected by fundamentals**: any $P_0 > (M_0/N_0)m_{ss}$ generates a hyperinflationary path. Nothing inside the model picks the store-of-value equilibrium over the continuum of shrinking-money paths.

Sims’s question. Once the government has a *fiscal side* (surpluses, taxes), do these bad equilibria survive?

Samuelson consumption-loan model with storage

Two-period endowment OLG, stationary, log utility. Young endowment w_1 ; old endowment $w_2 = 0$. Government issues nominal bonds B_t at gross nominal rate $1 + \theta$.

Households

$$\begin{aligned} \max \quad & \log(c_t^t) + \log(c_{t+1}^t) \\ \text{s.t.} \quad & c_t^t + s_t + \frac{B_t}{P_t} = w_1 \\ & c_{t+1}^t = \frac{B_t(1 + \theta)}{P_{t+1}} + \gamma s_t, \quad \gamma \in (0, 1) \end{aligned}$$

Government

$$B_{t+1} = (1 + \theta) B_t, \quad B_t \geq 0$$

Think $\theta = 0$ and $B \equiv M$: *paper money*.

Why storage?

Storage s_t at return $\gamma < 1$ is an **outside option** to holding government paper. In the classical Samuelson model the young *must* hold B/P to carry wealth forward. Storage breaks that monopoly: if government paper is expected to lose value, agents can coordinate on storage.

Optimality

First-order conditions:

$$\frac{c_{t+1}^t}{c_t^t} = \gamma \quad \text{if } s_t > 0, \quad \frac{c_{t+1}^t}{c_t^t} = (1 + \theta) \frac{P_t}{P_{t+1}} \quad \text{if } B_t > 0.$$

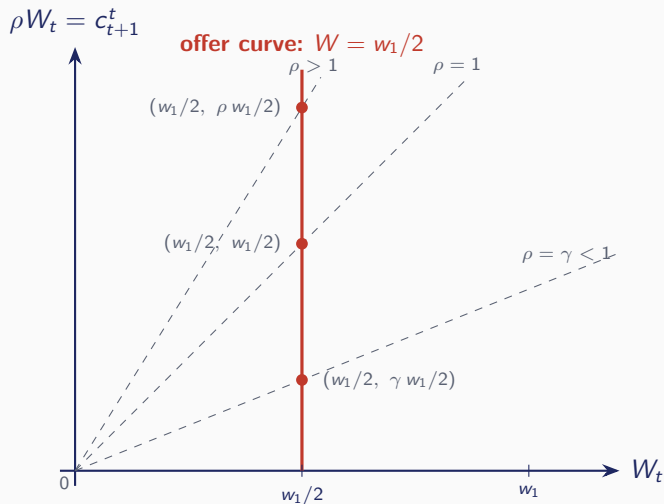
Let $W_t := s_t + \frac{B_t}{P_t}$ denote total savings and ρ_t the real return on W_t . Log utility $\Rightarrow$

$$\frac{c_{t+1}^t}{c_t^t} = \rho_t \implies W_t = c_t^t = \frac{w_1}{2}.$$

The offer curve is vertical

With $w_2 = 0$ and log utility, the young always save *exactly half* their endowment — independent of ρ_t . Income and substitution effects cancel. All adjustment between money and storage runs through the **composition of savings**, not its level. Why is the offer curve vertical?

Offer Curve



Equilibrium without storage: the monetary eqm

Savings is held entirely in bonds: $W_t = \frac{B_t}{P_t} = \frac{w_1}{2}$.

Goods-market clearing $c_t^t + c_t^{t-1} = w_1$ gives

$$\rho_t = 1, \quad 1 + \theta = \frac{P_{t+1}}{P_t}, \quad c_{t+1}^t = c_t^t = \frac{w_1}{2}.$$

Real debt is constant

$$B_{t+1} = (1 + \theta)B_t \Rightarrow B_{t+1} = (1 + \pi_{t+1})B_t \Rightarrow \frac{B_{t+1}}{P_{t+1}} = \frac{B_t}{P_t} = \frac{w_1}{2}$$

The initial-old consumption condition uniquely pins down the price level P_1

$$c_1^0 = (1 + \theta) \frac{B_0}{P_1} = \frac{w_1}{2} \rightarrow P_1 = \frac{2}{w_1} (1 + \theta) B_0$$

Intuition

Wallace (1998): when an alternative store of value (storage) is available, the use of money becomes an *endogenous* outcome rather than an assumption.

Equilibrium with storage: indeterminacy

No-arbitrage between storage and bonds: $\gamma = \rho_t = (1 + \theta) \frac{P_t}{P_{t+1}} < 1$. So inflation jumps to $P_{t+1}/P_t = (1 + \theta)/\gamma > (1 + \theta)$.

Plugging into the government BC:

$$\frac{B_t}{P_t} = \gamma \frac{B_{t-1}}{P_{t-1}} \implies \frac{B_t}{P_t} \rightarrow 0, \quad s_t \rightarrow \frac{w_1}{2}.$$

It is implicitly assumed that $s_1 \geq 0$ and $B_1 \geq 0$

$$W_1 = s_1 + \frac{B_1}{P_1} = \frac{w_1}{2} \rightarrow s_1 = \frac{w_1}{2} - \frac{B_1}{P_1} > 0 \rightarrow \frac{w_1}{2} - \frac{B_0(1 + \theta)}{P_1} > 0$$

Then any $P_1 > \frac{2}{w_1}(1 + \theta)B_0$ is an equilibrium: P_1 is indeterminate.

Welfare. Along any storage path, $c_t^{t-1} = \gamma \frac{w_1}{2} < \frac{w_1}{2}$ — strictly worse than the monetary equilibrium.

Equilibrium with storage: indeterminacy

Intuition

Storage equilibria are the **Sims analogue of the hyperinflationary paths**:

$$\underbrace{M/P_t \rightarrow 0}_{\text{classical OLG}} \longleftrightarrow \underbrace{B_t/P_t \rightarrow 0}_{\text{Sims, storage case}}$$

Both describe a coordination failure: agents expect government paper to lose value, shift wealth onto the alternative store of value, and vindicate their own expectations. OLG monetary indeterminacy has not gone away — it has just moved into the storage margin.

Adding a fiscal side: tax backing

Now assume the young pay a lump-sum tax $\tau > 0$.

Households

$$c_t^t + W_t + \tau = w_1$$

Government

$$\frac{B_t}{P_t} = (1 + \theta) \frac{B_{t-1}}{P_t} - \tau$$

From passive to active fiscal policy

In the pure-money model, the government budget is an *identity* that adjusts to whatever P_t agents coordinate on. With a rule $\tau > 0$ independent of debt, the budget becomes an **equation that constrains equilibrium prices**: surpluses must, in present value, equal the real value of outstanding debt. This is the FTPL logic.

Storage + tax backing: ruled out

Under no-arbitrage, $\rho_t = \gamma < 1$ still gives $\frac{B_t}{P_t} = \gamma \frac{B_{t-1}}{P_{t-1}} - \tau$. Iterating the government BC backwards:

$$\frac{B_t}{P_t} = \gamma^{t-1} \frac{B_1}{P_1} - \tau \sum_{j=0}^{t-1} \gamma^j \implies \lim_{t \rightarrow \infty} \frac{B_t}{P_t} = -\frac{\tau}{1-\gamma} < 0.$$

This violates $B_t \geq 0$ **in finite time**. Hence $\rho_t < 1$ is no longer an equilibrium.

Why backing kills the bad equilibria

Because τ is independent of the debt stock, persistent surpluses *exhaust* real debt: eventually the government would become a net saver, which the non-negativity of nominal debt rules out. Agents anticipate this and cannot coordinate on the shrinking-paper path. Demand for real savings rises at $t = 1$; P_1 falls; storage disappears.

No storage + tax backing: the FTPL equilibrium

Effective young endowment is $w_1 - \tau$, so

$$W_t = \frac{w_1 - \tau}{2}, \quad c_t^t = \frac{w_1 - \tau}{2}, \quad c_{t+1}^t = \frac{w_1 + \tau}{2} \rightarrow \rho_t = \frac{w_1 + \tau}{w_1 - \tau} > 1$$

The government BC is $\frac{B_t}{P_t} = \frac{B_{t-1}}{P_{t-1}} \rho - \tau$ has a constant real-debt solution:

Debt valuation (FTPL) equation

$$\frac{B_t}{P_t} = \frac{\tau}{\rho - 1} \iff \text{real debt} = \text{PV of surpluses.}$$

The selection mechanism

The real value of government liabilities is pinned down by the PV of taxes. Via asset-market clearing $\frac{B_t}{P_t} = \frac{w_1 - \tau}{2}$, this in turn **determines P_1 uniquely** — no belief coordination needed. The FTPL is a fiscal theory of *equilibrium selection*, not a separate monetary theory.

Taking stock

Without fiscal backing

- 1 eqm without storage: paper valued,
 $1 = (1 + \theta) P_t / P_{t+1}$
- ∞ eqa with storage: $\frac{B_t}{P_t} \rightarrow 0$, P_1
indeterminate

Fragility: valued paper requires belief coordination.

With fiscal backing

- storage paths *ruled out* by $B_t \geq 0$ and persistent surpluses
- **unique eqm** determined by $\frac{B_t}{P_t} = \tau / (\rho - 1)$
- welfare below Samuelson eqm, but arbitrarily close as $\tau \rightarrow 0$

Fiscal policy replaces coordinated beliefs as the selection device.

The FTPL reading of classical OLG

What the OLG monetary literature treated as a deep *belief-coordination* problem, Sims re-reads as a *fiscal-regime* problem. Which equilibrium you are in depends on whether the government's surpluses respond to debt ("monetary dominance") or not ("fiscal dominance" / FTPL).

How robust is this? Hagedorn's critique (2024-2025)

Looking back at Sims's derivation, the uniqueness result rests on **two assumptions** used jointly:

Assumption 1

$$w_2 = 0$$

No old-age endowment. Drives...

- the tractable saving rule $W_t = w_1/2$;
- the *vertical* offer curve;
- the ρ -independence of young consumption.

Assumption 2

$$B_t \geq 0$$

Non-negative debt. Used to...

- rule out storage paths where the government would become a net saver;
- force the FTPL equilibrium with $\rho > 1$ as the *only* survivor.

Hagedorn's claim

Each assumption looks innocuous on its own. **Jointly** they rule out by assumption the very multiplicity that would otherwise arise. Relax either one, and classical OLG indeterminacy resurfaces — fiscal backing alone does *not* deliver uniqueness → FTPL generally fails in OLG.

Generalising the household problem

Replace Sims's assumptions with:

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad w_2 > 0, \quad B_t \in \mathbb{R}.$$

Household problem (net of taxes τ):

$$\max u(c_t^t) + u(c_{t+1}^t) \quad \text{s.t.} \quad c_t^t = w_1 - \tau - W_t, \quad c_{t+1}^t = w_2 + \rho W_t.$$

Euler equation: $u'(c_t^t) = \rho u'(c_{t+1}^t) \iff (c_t^t)^{-\sigma} = \rho (c_{t+1}^t)^{-\sigma}.$

Solving for W_t gives the **general saving function**:

$$W(\rho) = \frac{w_1 - \tau - [\rho]^{-\sigma} w_2}{1 + \rho^{1-\sigma}}$$

Sanity check. Set $w_2 = 0$, $\sigma = 1$: giving $W(\rho) = (w_1 - \tau)/2$. **Back to Sims's flat saving rule.**

Steady-state equilibrium: two curves

Savings demand

Household wants to save $W(\rho)$ for *any* given return:

$$W(\rho) = \frac{w_1 - \tau - [\rho]^{-\sigma} w_2}{1 + \rho^{1-\sigma}}.$$

Debt supply (FTPL)

Government BC in steady state: $\frac{B_t}{P_t} = \rho \frac{B_t}{P_t} - \tau \Rightarrow$

$$\frac{B_t}{P_t} = \frac{\tau}{\rho - 1}.$$

Same equation as before, just written for the *steady state*.

Asset-market clearing: households must actually hold the debt the government issues.

$$W(\rho) = \frac{B_t}{P_t} \quad \Longleftrightarrow \quad \boxed{\Gamma(\rho) := (\rho - 1) W(\rho) - \tau = 0}$$

What we are looking for

Steady-state real returns ρ satisfying $\Gamma(\rho) = 0$. In Sims's knife-edge case this equation has a unique positive root. In the general case it has **more than one**.

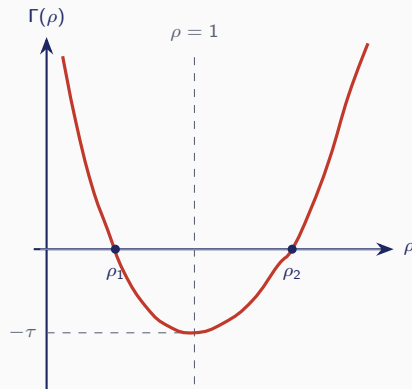
Boundary behaviour of $\Gamma(\rho)$

Evaluate Γ at three limits:

- $\rho \rightarrow 0^+$: $W(\rho) \rightarrow -\infty$ (because $[\rho]^{-\sigma} w_2$ blows up in the numerator when $w_2 > 0$)
 $\Rightarrow \Gamma(\rho) = (\rho - 1)W(\rho) - \tau \rightarrow +\infty$.
- $\rho = 1$: the factor $(\rho - 1)$ vanishes, so
 $\Gamma(1) = 0 \cdot W(1) - \tau = -\tau < 0$.
- $\rho \rightarrow \infty$: debt supply collapses, $W(\rho)$ stays bounded,
 $\Rightarrow \Gamma(\rho) \rightarrow +\infty$.

IVT $\Rightarrow$ at least two roots

$$\rho_1 \in (0, 1), \quad \rho_2 \in (1, \infty).$$



The two steady states: economic content

$\rho_2 > 1$: Sims's FTPL

- Positive real interest rate.
- $\frac{B_t}{P_t} = \tau/(\rho_2 - 1) > 0$: government is a *net debtor*.
- Dynamically efficient.
- The equilibrium Sims analyses.

$\rho_1 < 1$: the “bad” root

- Real return below 1: dynamically *inefficient*.
- $\frac{B_t}{P_t} = \tau/(\rho_1 - 1) < 0$: government is a *net creditor* / households borrow.
- This is what $B_t \geq 0$ was ruling out!

Why Sims's non-negativity restriction is now ad-hoc

With $w_2 = 0$, households have no income in old age and therefore cannot meaningfully borrow — the constraint $B_t \geq 0$ binds vacuously. With $w_2 > 0$, households **have future income to borrow against**, so negative-debt equilibria are economically sensible. Ruling them out by fiat is analogous to selecting the “good” bubble equilibrium by assumption: it works, but offers no deeper justification.